

Utility-Driven Coordinated Multi-UAV Exploration with Target Detection and Localization

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Repository: <https://github.com/i-sonu/ai-multi-uav-swarm-searcher>

1. Problem Identification

1.1 Application Domain

This project addresses autonomous aerial exploration of unknown environments for post-disaster search and reconnaissance. The task is to have a small team of unmanned aerial vehicles enter an area for which no prior map exists, build a map of it cooperatively, and simultaneously record the location of targets of interest (in the motivating scenario, people) so that limited human rescue effort can be directed to where it is most likely to matter.

The scale of the underlying problem is well documented. The EM-DAT annual disaster report for 2024 records hundreds of disaster events worldwide with economic damages on the order of hundreds of billions of US dollars, including droughts in Africa affecting over 25 million people. Beyond scale, the defining characteristic of the domain is time. Emergency response literature consistently identifies the first 72 hours after a structural collapse as the "golden period": survival probability is highest in the first 24 hours and falls steeply thereafter, with rescue experts noting that the large majority of earthquake survivors are recovered within the first three days. Search effort expended in the wrong place during that window is effectively irrecoverable.

Unmanned aerial vehicles are increasingly used in this setting because they can survey large and inaccessible areas rapidly. However, their effectiveness depends on two capabilities that are usually developed in isolation: deciding where to fly next in an unmapped environment, and reliably recognising small, sparse human figures in aerial imagery. Combining the two (letting what the vehicle detects influence where it searches next, and recording where each detection occurred) is the gap this project targets.

1.2 Stakeholder and Supported Decision

The stakeholder is the incident commander of a search-and-rescue or disaster-response team, together with the UAV operators reporting to them. The decision the system supports is resource-directing: given a bounded but unmapped search area and a limited number of ground rescue teams, which regions have already been searched, which remain unsearched, and where have candidate targets been observed? The system's output is intended to convert an unstructured search area into a prioritised, spatially indexed picture of what is known and what has been found, so that the commander can allocate ground teams on evidence rather than on guesswork.

A secondary stakeholder framing, useful for evaluation, is the autonomy researcher: the same system serves as a testbed for comparing coordination strategies in multi-agent exploration under controlled and repeatable conditions.

2. Literature Survey

The survey below covers three converging strands: (i) classical frontier-based exploration and its multi-robot extensions, (ii) recent multi-UAV and learning-based cooperative exploration, and (iii) aerial target detection for search and rescue. Foundational works are retained because they define the baselines against which this project is evaluated; the majority of entries are from the last three years.

Ref.	Year	Dataset / Environment	Method / Architecture	Key Metric & Value	Stated Limitation
[1]	1997	Simulated + real indoor occupancy grids	Frontier-based exploration; robot drives to nearest boundary between free and unknown cells	Demonstrated full coverage of office environments (foundational, pre-benchmark era)	Single robot only; greedy nearest-frontier choice is myopic and revisits regions
[2]	1998	Multi-robot indoor simulation	Extension of frontier exploration to multiple robots sharing a map	Reduced exploration time vs. single robot	No explicit coordination — robots may select the same frontier, causing redundant coverage
[3]	2005	Indoor office environments, real robot teams	Centralised coordinated exploration; utility-cost trade-off assigns robots to distinct frontiers	Significant reduction in total exploration time vs. uncoordinated baseline	Centralised: heavy communication and computation cost; poor scaling with team size
[4]	2016	3D indoor/outdoor simulation	Receding-horizon Next-Best-View (NBV) planner using RRT sampling in 3D	Enabled 3D volumetric exploration with onboard compute	Sampling-based NBV is computationally heavy; single-agent formulation
[5]	2020	3-D indoor environments	Incrementally built topological map for autonomous exploration	Improved exploration efficiency vs. dense grid methods (IEEE T-IM)	Topological abstraction can miss fine structure; single-agent
[6]	2022	Disaster-response task scenarios	Multi-robot task allocation with deadlines, range and payload constraints	Improved task completion under deadline constraints	Assumes known task locations — not coupled to online exploration
[7]	2022	Subterranean / intermittent-connectivity settings	ACHORD: communication-aware multi-robot coordination	Maintained coordination under intermittent connectivity (IEEE RA-L)	Substantial infrastructure assumptions; complex comms stack
[8]	2023	Forest environments, multi-UAV	Fast decentralised multi-UAV exploration	Fast exploration in cluttered forest (IEEE RA-L 8(9))	Decentralised design assumes reliable local sensing; heavy onboard compute
[9]	2023	Terrestrial simulation environments	Cooperative autonomous exploration via explicit task allocation	Reduced overlap vs. uncoordinated frontier baseline	Ground robots only; allocation not evaluated under sensing noise
[10]	2024	Multi-robot simulation benchmarks	Transformer-based reinforcement learning for autonomous exploration	Outperformed classical frontier baselines (Sensors 24(16):5083)	Requires large-scale training; sim-to-real transfer unvalidated
[11]	2024	Distributed multi-UAV, bandwidth-limited	LECES: low-bandwidth collaborative exploration system	Efficient collaboration at reduced bandwidth (IEEE RA-L 9(9))	Optimised for comms efficiency rather than exploration optimality
[12]	2024	Indoor drone imagery	YOLO-IHD: improved real-time human detection for indoor drones	Real-time human detection on drone hardware (Sensors 24:922)	Detection only — no coupling to exploration or mapping

Ref.	Year	Dataset / Environment	Method / Architecture	Key Metric & Value	Stated Limitation
[13]	2025	VisDrone (pre-train) + Heridal (fine-tune)	YOLOv5s with PB-FPN and deconvolution; two-stage transfer learning	mAP@50 = 0.802 on Heridal, real-time on Jetson Nano	Single-modality RGB; no integration with autonomous flight planning
[14]	2025	Survey of aerial SAR person-detection benchmarks	Survey and benchmark of aerial person detection for search and rescue	Consolidated benchmark comparison (J. Remote Sensing 5:0474)	Survey — highlights that detection is evaluated offline, decoupled from mission planning
[15]	2025	Bandwidth-limited multi-UAV simulation	PC-Explorer: decentralised multi-UAV exploration	Exploration under bandwidth limits (IEEE RA-L 10(9))	Focus on communication constraints; no semantic/target-aware objective
[16]	2025	Communication-constrained multi-robot settings	Collaborative exploration under communication constraints	Improved coverage under constrained comms (IEEE Access 13)	Geometric coverage objective only — no target-detection reward
[17]	2025	Multi-robot simulation	Deep RL with knowledge distillation for collaborative exploration	Improved efficiency vs. non-distilled RL (Mathematics 13(1):173)	Training cost high; generalisation to unseen maps limited
[18]	2025	Systematic review, 2013–2024 corpus	Systematic literature review on multi-robot task allocation	Taxonomy of MRTA methods (ACM Comput. Surv.)	Review notes most MRTA work assumes tasks are known a priori, not discovered online

2.1 Research Gap

Five gaps emerge from the comparison above and define the space this project occupies.

- **1. Exploration and detection are optimised separately.** Detection work [12], [13], [14] evaluates models offline on fixed aerial datasets and reports mAP, with no coupling to a flight-planning policy. Exploration work [8], [11], [15], [16] optimises purely geometric coverage and is indifferent to what is observed. Almost no work closes the loop so that detections are spatially registered into the same map that drives the next exploration decision.
- **2. Coordination benefit is rarely isolated as a controlled variable.** Multi-agent systems are typically compared against other multi-agent systems, or reported only as aggregate exploration time. The specific marginal contribution of the allocation mechanism — coordinated versus uncoordinated agents holding team size, sensing and planner constant — is seldom reported as a clean ablation.
- **3. Recent effort concentrates on communication constraints rather than allocation quality.** Several 2022-2025 systems [7], [11], [15], [16] are motivated primarily by bandwidth and connectivity limits. This is important for field deployment but orthogonal to the question of how frontiers should be assigned when communication is adequate, which remains comparatively under-examined at small team sizes.
- **4. Learning-based exploration lacks transparent classical baselines.** Reinforcement-learning approaches [10], [17] report improvements over frontier baselines but carry high training cost and limited generalisation to unseen maps, and the classical baselines they compare against are often under-specified.
- **5. Task allocation literature assumes tasks are known in advance.** The systematic review of multi-robot task allocation [18] observes that most formulations assume a known task set. In exploration, the "tasks" (frontiers) are generated and destroyed continuously by the agents'

own sensing, which makes the allocation problem non-stationary in a way standard MRTA formulations do not address.

3. Problem Statement

Given synchronous 2-D LiDAR range scans and downward-facing RGB imagery from two simulated UAVs deployed into a bounded 60 m x 60 m environment for which no prior map exists, discretised into a shared 240 x 240 occupancy grid at 0.25 m resolution, produce (a) a complete free/occupied/unknown labelling of the environment and (b) a georeferenced register of detected targets with grid coordinates, class label, confidence and first-detection provenance; subject to the constraints that all decisions are made online without a prior map, that the two agents must not redundantly cover the same frontier region, that frontier assignment must complete within the inter-decision interval on commodity hardware, and that targets are small, sparse and class-imbalanced in aerial imagery with limited annotated training data, such that time-to-90%-coverage is reduced by at least 35% relative to a single-agent nearest-frontier baseline and the redundant-coverage ratio is measurably lower than that of a two-agent uncoordinated nearest-frontier baseline, with detection quality reported as mAP@50 on a held-out, scale-stratified test split.

The four required elements of the statement are itemised below for clarity.

Element	Specification in this work
Input	Per-agent 2-D LiDAR range scans (360 beams, 12 m max range, 10 Hz) and a downward-facing RGB camera stream (640 x 480, 5 Hz) from two simulated UAVs operating in a bounded unknown environment of approximately 60 m x 60 m, discretised to a 0.25 m occupancy grid (240 x 240 cells).
Output	A shared global occupancy grid labelling every cell free / occupied / unknown, plus a georeferenced target register: a list of detected targets with grid coordinates, class label and confidence, and the timestamp and agent ID of first detection.
Constraint	Decisions must be made online with no prior map; the two agents must avoid redundant coverage of the same frontier; allocation must run within the inter-decision interval on commodity hardware; and the detector operates on small, sparse aerial targets under class imbalance with a limited annotated training set.
Success criterion	Time-to-90%-coverage and total path length, benchmarked against (i) a single-agent nearest-frontier baseline and (ii) a two-agent uncoordinated (independent nearest-frontier) baseline, targeting at least a 35% reduction in time-to-90%-coverage over the single-agent baseline and a measurable reduction in redundant-coverage ratio over the uncoordinated baseline; detection reported as mAP@50 on a held-out split.

4. Proposed System Architecture

4.1 Pipeline Overview

The system is organised as a closed perception-planning-action loop replicated per agent, with a single shared world model mediating between them. The shared model is what makes the two agents a team rather than two independent explorers.

Figure 1 — System Pipeline: Coordinated Two-Agent Aerial Exploration with Online Target Localisation

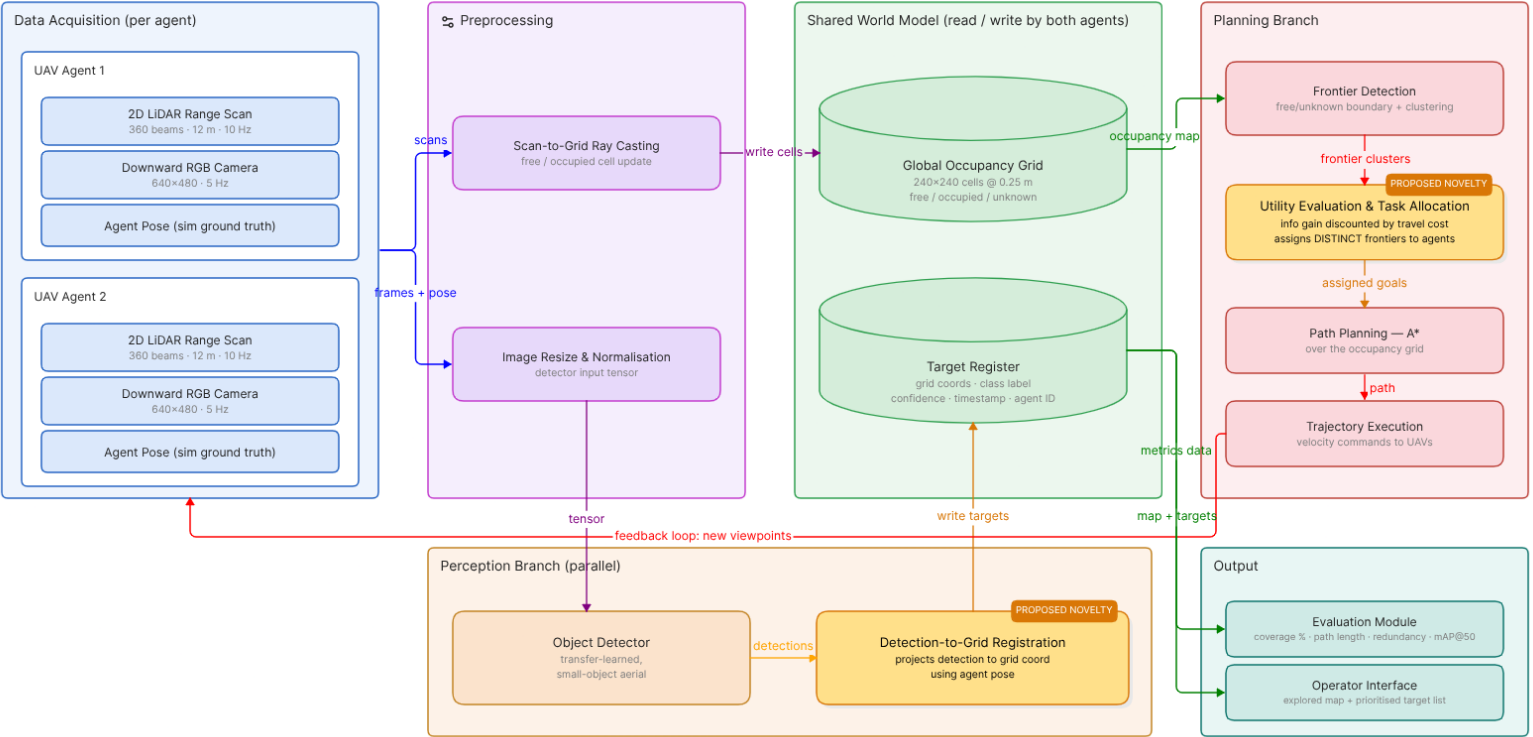


Figure 1. End-to-end system pipeline from data acquisition through to operator output.

The loop proceeds as follows. Each agent's range scan is ray-cast into the shared occupancy grid, updating cells to free or occupied. Frontier detection then extracts the boundary cells between free and unknown space and clusters them into candidate goals. The allocation module scores each candidate frontier for each agent and assigns distinct frontiers to the two agents. Each agent plans a path to its assigned frontier with A* over the current grid and begins executing it, replanning when the map changes materially. In parallel, RGB frames are passed to the detector; any detection is projected to a grid coordinate using the agent's pose and written to the target register, which is part of the same shared model.

4.2 Detailed Module Diagram

The second diagram expands the two components carrying the project's contribution — the allocation module and the detector — with explicit input and output shapes.

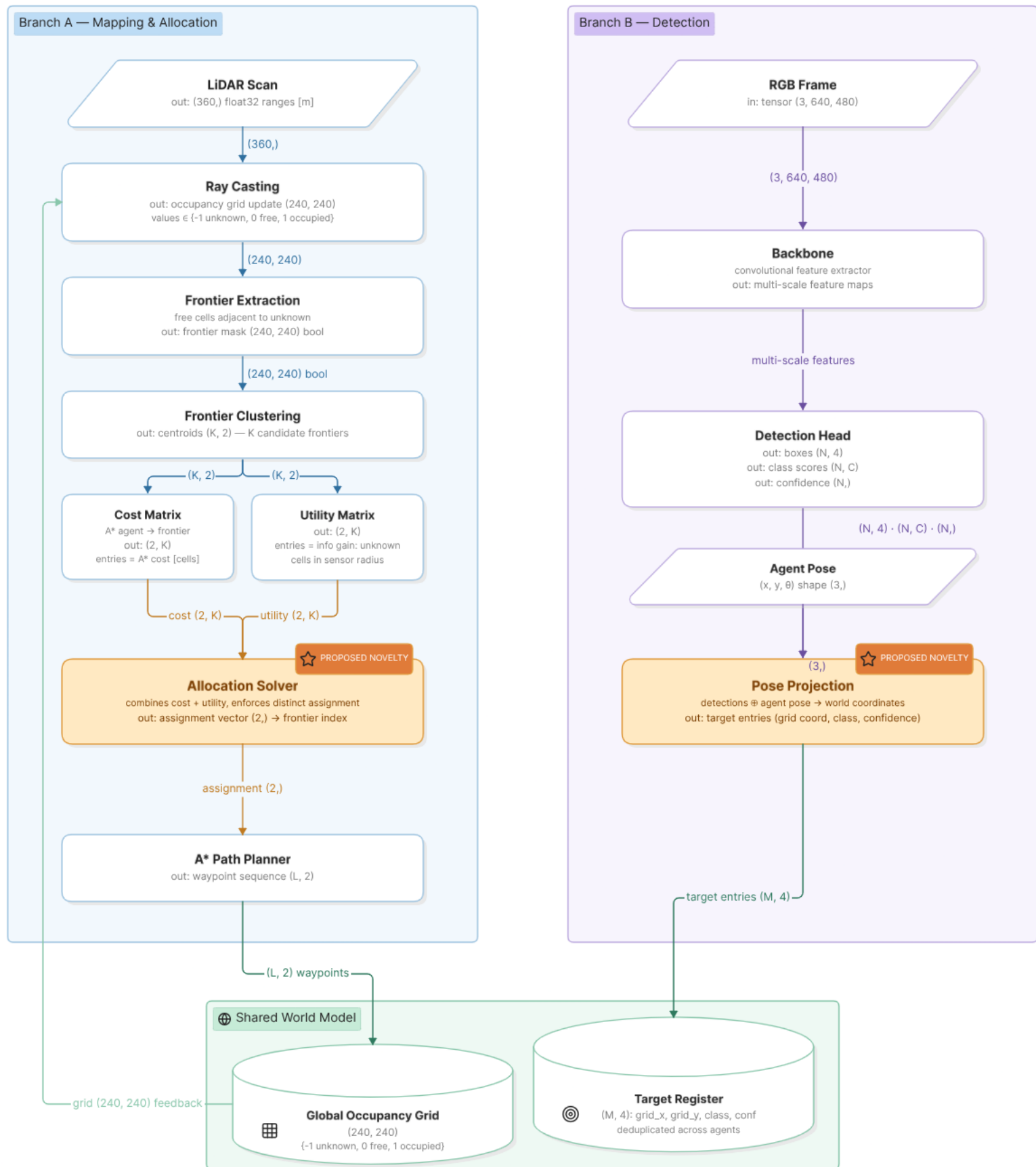


Figure 2. Detailed module diagram. Shaded blocks indicate the proposed contribution.

Novelty placement. The contribution is not a new detector architecture or a new path-planning algorithm. It is located precisely at two points: (i) the utility-based allocation block, which assigns frontiers to agents using a combined distance and information-gain score rather than independent greedy selection, and (ii) the detection-to-grid registration block, which writes detections into the

same shared spatial structure that drives exploration, making the target register a first-class output rather than a post-hoc log.

4.3 Justification of Design Choices

- **Frontier-based exploration as the substrate.** Frontier exploration [1] is chosen over sampling-based next-best-view planning [4] because it is deterministic, cheap to compute on a 2-D grid, and directly interpretable — properties that matter for a system whose purpose includes isolating the effect of coordination. Sampling-based 3-D planners are computationally heavier and introduce stochasticity that would confound the allocation ablation.
- **A* for frontier-reaching paths.** A* with an admissible heuristic returns optimal paths while expanding fewer nodes than uniform-cost search. Because path cost is the input to the allocation cost matrix, the quality of the assignment depends directly on the accuracy of these costs; using an optimal planner ensures the allocation is not being fed misleading estimates. BFS, DFS and UCS are retained as measured baselines rather than discarded, since the comparison is itself a reportable result.
- **Centralised shared map with ground-truth pose.** Centralised coordination is known to incur communication and computation cost and to scale poorly with team size [3]. At a team size of two, however, that cost is negligible, and centralisation removes map-merging and localisation drift as confounding variables. This is a deliberate scoping decision: the hypothesis under test concerns allocation quality, and decentralisation [8], [15] addresses a different question.
- **Utility-based rather than purely distance-based allocation.** Uncoordinated nearest-frontier selection allows both agents to converge on the same region, which is the limitation explicitly noted in the earliest multi-robot frontier work [2]. Scoring frontiers by expected information gain discounted by travel cost, and enforcing distinct assignments, is hypothesised to reduce the redundant-coverage ratio. This is stated as a hypothesis to be tested by ablation A1, not assumed.
- **Transfer learning for the detector.** Two-stage transfer learning from a general aerial dataset followed by task-specific fine-tuning has been shown effective for small-object aerial search and rescue detection, reaching mAP@50 of 0.802 on the Heridal benchmark while remaining real-time on embedded hardware [13]. Given the limited annotated data available to this project, training from scratch is not viable and transfer learning is the evidence-supported alternative.
- **Two-dimensional occupancy mapping.** Restricting to a 2-D grid at fixed flight altitude removes 3-D volumetric mapping and full SLAM from the critical path. This is a feasibility decision, made so that the algorithmic contributions can be completed and evaluated rigorously rather than partially demonstrated.

4.4 Planned Experimental Setup

Item	Plan
Environment / dataset	Exploration: procedurally generated 2-D maps (office, maze, open-field, cluttered) in a custom Python simulator, plus Gazebo worlds for the ROS 2 phase. Detection: VisDrone publicly available aerial person-detection dataset for pre-training, and rendered synthetic aerial frames captured from the simulator for domain-specific fine-tuning.
Split strategy	Exploration: 30 held-out map seeds per environment class, never used during parameter tuning. Detection: standard train/val/test split, stratified by target scale so small-object performance is separable.
Evaluation metrics	Exploration: time-to-90%-coverage, total and per-agent path length, redundant-coverage ratio (fraction of cells observed more than once), nodes expanded and wall-clock planning time per decision. Detection: mAP@50, mAP@50-95,

Item	Plan
	precision/recall on the small-object subset. Combined: time-to-first-detection and fraction of targets localised within a 1 m grid tolerance.
Baselines	B1: single agent, nearest-frontier. B2: two agents, uncoordinated nearest-frontier (no allocation). B3: two agents, greedy sequential allocation. Proposed: two agents with utility-based / auction allocation. Path-planning baselines: BFS, DFS and Uniform-Cost Search against A*.
Hardware	Development on a consumer laptop / lab workstation for the 2-D simulator (CPU-only, sufficient for the search algorithms). Detector training on an available CUDA GPU or a free-tier cloud GPU session; inference benchmarked on the target edge device where available.
Ablations	A1: allocation strategy (none / greedy / utility-based auction). A2: informed vs. uninformed search for the frontier-reaching path (A* vs. UCS vs. BFS). A3: frontier utility weighting (pure distance vs. distance and information-gain). A4: team size (1 vs. 2 agents) at matched total flight time. A5: detector backbone size vs. inference latency.

5. Feasibility Note

Compute. The exploration component is CPU-bound and runs comfortably on a standard laptop: a single episode on a 240 x 240 grid completes in seconds, which makes the full ablation grid (five studies across thirty held-out map seeds and four environment classes) tractable to run repeatedly rather than once. The detector is the only GPU-dependent component; fine-tuning a small backbone at reduced input resolution is estimated at a few GPU-hours, which is within reach of an available CUDA workstation or free-tier cloud GPU sessions.

Data. Exploration data is generated procedurally, so dataset access is not a risk and the volume is unbounded. For detection, public aerial person-detection benchmarks referenced in the survey literature [13], [14] are openly available for pre-training, supplemented by synthetic aerial frames rendered from the simulator, for which ground-truth labels are obtained automatically from the simulator state. Access status for the public datasets is to be confirmed and recorded in the repository.

Staging. The work is deliberately phased so that a demonstrable system exists at every review point. Phase 1 delivers a lightweight 2-D simulator with occupancy mapping, frontier detection and A* — sufficient on its own to produce the search-algorithm comparison. Phase 2 adds the second agent and the allocation mechanism, which is the central contribution. Phase 3 adds the trained detector and target registration, and ports the validated algorithms to a higher-fidelity ROS 2 / Gazebo environment. Each phase is independently demonstrable, so a delay in a later phase degrades scope rather than producing nothing.

Risks and fallbacks. The principal identified risks and their mitigations are tabulated below.

Risk	Impact	Fallback plan
ROS 2 / Gazebo environment setup and dependency issues consume more time than budgeted	Delays the high-fidelity simulation phase	The lightweight 2-D simulator is developed first and is independently sufficient to produce every algorithmic result and ablation in this report; the Gazebo port is an enhancement, not a dependency
Multi-agent allocation does not outperform the uncoordinated	Weakens the central claim	Report the negative result honestly with per-map-class

Risk	Impact	Fallback plan
baseline on some map classes		analysis; the comparison itself remains a valid contribution, and map geometry is added as an explicit experimental factor
Insufficient annotated aerial data for the detector, or poor small-object performance	Detection component underperforms	Fall back to transfer learning from a public aerial dataset with synthetic frames rendered from the simulator; if accuracy remains low, report detection as a component study and keep exploration as the primary contribution
GPU access unavailable for detector training	Blocks the learning component	Use a smaller backbone at reduced input resolution, free-tier cloud GPU sessions, or a frozen pre-trained backbone with only the detection head fine-tuned
Scope creep from adding 3-D exploration, SLAM, or more than two agents	Nothing finishes to a demonstrable standard	Scope is explicitly frozen at 2-D occupancy mapping with ground-truth pose and a two-agent team; extensions are recorded as future work only

6. Contribution Matrix

Both members participate in all major design decisions. "Primary Owner" indicates responsibility for execution and for presenting that component at review; "Secondary Owner" reviews, tests and supports it. Effort is split 50/50.

SaiAmirthesh — Lead AI Systems & Search Architect: core AI algorithm design, graph search implementations, multi-agent dynamic task allocation (Hungarian and greedy scoring), computer vision detection pipeline, target spatial registration, and statistical experiment design.

Adarsh — Swarm Systems, Benchmarking & Infrastructure Lead: simulation environment generation, LiDAR sensor ray-casting mechanics, multi-agent observation tracking, headless parallel experiment execution framework, rendering and visualisation tools, ROS 2 / Gazebo bridge, and the automated testing suite.

WBS	Phase / Task Description	Primary Owner	Secondary Owner	Deliverables / Artifacts	Status
0.0	Project Planning & Research Foundation				Completed
0.1	Project Identification & Problem Definition	Adarsh	SaiAmirthesh		Completed
0.2	Literature Survey & Related Work Analysis	Adarsh	SaiAmirthesh		Completed
0.3	Research Gap Identification & Novelty Definition	Adarsh	SaiAmirthesh		Completed
0.4	Functional & Non-Functional Requirement Analysis	SaiAmirthesh	Adarsh		Completed
0.5	High-Level System Architecture & Module Design	Adarsh	SaiAmirthesh		Completed
0.6	Algorithm Selection & Technology Stack Evaluation	SaiAmirthesh	Adarsh		Completed
0.7	Dataset & Simulation Environment Planning	SaiAmirthesh	Adarsh		Completed
0.8	Work Breakdown Structure (WBS) & Development Roadmap	SaiAmirthesh	Adarsh		Completed
0.9	GitHub Repository, Coding Standards & Project Setup	Adarsh	SaiAmirthesh		Completed

WBS	Phase / Task Description	Primary Owner	Secondary Owner	Deliverables / Artifacts	Status
1.0	Environment & Single-Agent Foundation				Completed
1.1	Occupancy Map Generators (Office, Maze, Open Field, Obstacles)	Adarsh	SaiAmirthesh	src/world/map_generator.py	Completed
1.2	Ray-casting LiDAR Sensor Model & Noise Simulation	Adarsh	SaiAmirthesh	src/world/sensor.py	Completed
1.3	Shared Occupancy Grid World Model & State Tracking	SaiAmirthesh	Adarsh	src/mapping/occupancy_grid.py	Completed
1.4	Single-Agent Pose & Kinematic Motion Simulator	SaiAmirthesh	Adarsh	src/agents/agent.py	Completed
1.5	Phase 1 Fixed-Patrol Demo & Live Visualizer	Adarsh	SaiAmirthesh	scripts/demo_phase1.py	Completed
2.0	Single-Agent Autonomous Exploration & Path Planning				Completed
2.1	Graph Search Algorithms (A*, BFS, DFS, UCS)	Adarsh	SaiAmirthesh	src/planning/astar.py, baselines.py	Completed
2.2	Frontier Boundary Detection Algorithm	Adarsh	SaiAmirthesh	src/frontier/detection.py	Completed
2.3	Frontier Clustering via Distance-Based DBSCAN	Adarsh	SaiAmirthesh	src/frontier/clustering.py	Completed
2.4	Single-Agent Autonomous Exploration Loop & Blacklisting	Adarsh	SaiAmirthesh	src/exploration/explorer.py	Completed
2.5	Phase 2 Autonomous Exploration Demo Script	Adarsh	SaiAmirthesh	scripts/demo_phase2.py	Completed
3.0	Metrics Engine & Baseline Benchmarking (Experiment E1)				Completed
3.1	Core Metrics Module (Coverage %, Path Length, Time, Expansions)	Adarsh	SaiAmirthesh	src/metrics/metrics.py	Completed
3.2	Headless Parallel Experiment Runner	Adarsh	SaiAmirthesh	src/experiments/runner.py	Completed
3.3	Held-out Test Seeds Dataset Configuration	Adarsh	SaiAmirthesh	configs/experiments/heldout_seeds.yaml	Completed
3.4	Experiment E1 Execution (480 held-out runs across planners)	Adarsh	SaiAmirthesh	scripts/run_e1.py	Completed
3.5	Publication-Quality Figure Generation Scripts	Adarsh	SaiAmirthesh	scripts/plot_e1.py	Completed
3.6	Phase 3 Written Report & Empirical Findings	Adarsh	SaiAmirthesh	docs/results_e1.md	Completed
4.0	Multi-UAV Swarm Coordination & Task Allocation				In Progress
4.1	N-Agent Infrastructure & Shared Map Access Synchronization	SaiAmirthesh	Adarsh	src/agents/agent.py, explorer.py	Completed
4.2	Per-Agent Cell Observation & Redundant Coverage Tracker	SaiAmirthesh	Adarsh	src/metrics/metrics.py	Completed
4.3	Baseline B2 Implementation (Uncoordinated Nearest-Frontier)	SaiAmirthesh	Adarsh	src/planning/allocation.py	Completed
4.4a	Cost-Utility Matrix Builder (A* Cost + Info Gain Utility)	SaiAmirthesh	Adarsh	src/planning/allocation.py	Completed
4.4b	Greedy Multi-Agent Allocator	SaiAmirthesh	Adarsh	src/planning/allocation.py	Completed
4.4c	Optimal Hungarian Assignment Allocator (linear_sum_assignment)	SaiAmirthesh	Adarsh	src/planning/allocation.py	Completed
4.4d	Dynamic Re-planning & Goal Invalidated Re-assignment Policy	SaiAmirthesh	Adarsh	src/planning/allocation.py	Completed
4.5	Experiment E2 (Allocation Ablation: Uncoordinated vs Coordinated)	SaiAmirthesh	Adarsh	scripts/run_e2.py	Completed
4.6	Experiment E3 (Team Size Scaling: N=1, 2, 3 at matched flight time)	SaiAmirthesh	Adarsh	scripts/run_e3.py	Completed
4.7	Experiment E4 (Cost-Utility Weight λ Sensitivity Sweep)	SaiAmirthesh	Adarsh	scripts/run_e4.py	Completed
4.8	Phase 4 Multi-Agent Benchmark Write-up	SaiAmirthesh	Adarsh	docs/results_e2_e4.md	Completed
5.0	Perception, Aerial Target Detection & Spatial Registration				Planned

WBS	Phase / Task Description	Primary Owner	Secondary Owner	Deliverables / Artifacts	Status
5.1	Aerial Person Detection Dataset Ingestion & Preprocessing	SaiAmirthesh	Adarsh	src/perception/data_loader.py	Completed
5.2	Object Detector Fine-Tuning (YOLOv8 / MobileNet-SSD)	SaiAmirthesh	Adarsh	src/perception/train.py	Pending
5.3	Detector Evaluation (mAP@50, mAP@50-95, Small-Object Subset)	SaiAmirthesh	Adarsh	src/perception/eval.py	Pending
5.4	Target Placement & Sensor Camera Footprint Simulation	Adarsh	SaiAmirthesh	src/world/sensor.py, agent.py	Pending
5.5	Spatial Target Registration Engine (TargetRegister & Deduplication)	SaiAmirthesh	Adarsh	src/perception/registration.py	Pending
5.6	Experiment E5 (Exploration Strategy vs Target Detection Rate)	SaiAmirthesh	Adarsh	scripts/run_e5.py	Pending
5.7	Visual Target Pinning Overlay on Rendering Engine	Adarsh	SaiAmirthesh	src/viz/render.py	Pending
6.0	ROS 2 & Gazebo High-Fidelity Simulation Bridge (Optional)				Pending
6.1	ROS 2 Package Architecture & Workspace Configuration	Adarsh	SaiAmirthesh	ros2_ws/ setup	Pending
6.2	Dual-Drone Gazebo Model Spawning & Waypoint Control Nodes	Adarsh	SaiAmirthesh	ros2_ws/src/uav_gazebo	Pending
6.3	Multi-Agent Occupancy Map Aggregator & ROS 2 Services	SaiAmirthesh	Adarsh	ros2_ws/src/uav_planner	Pending
6.4	Gazebo Exploration Video Recording & Demonstration	Adarsh	SaiAmirthesh	results/videos/gazebo_demo.mp4	Pending
7.0	Reproducibility, Verification & Master Documentation				Final Stage
7.1	Automated Test Suite & Regression Checks	Adarsh	SaiAmirthesh	scripts/test.sh, tests/	In Progress
7.2	Master Experiment Results Consolidation	SaiAmirthesh	Adarsh	docs/RESULTS.md	Pending
7.3	Final AI Technical README	SaiAmirthesh	Adarsh	docs/README.md	Completed
7.4	Final Codebase Verification & Citations Updating	Adarsh	SaiAmirthesh	CITATIONS.md, README.md	Pending

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